

EE168 Lab 2 Report

Layout Design, DRC, NAND, Ring Oscillator

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Summary

This lab walked me through the full CMOS layout flow, starting from a basic inverter and working up to a NAND gate and a hierarchical ring oscillator. I practiced drawing layouts, checking rule violations, and running DRC multiple times until everything passed. I also built simple testbenches, ran DC and transient simulations, and compared the pre-layout and post-layout waveforms to see how parasitics affect delay and signal shape. The ring oscillator part especially helped me understand device sizing, hierarchy, and how small routing choices change behavior. Overall, this lab made me a lot more confident in taking a schematic to a clean layout and verifying its functionality from the waveforms.

1 Inverter Layout, DRC, LVS

1.1 Inverter Layout

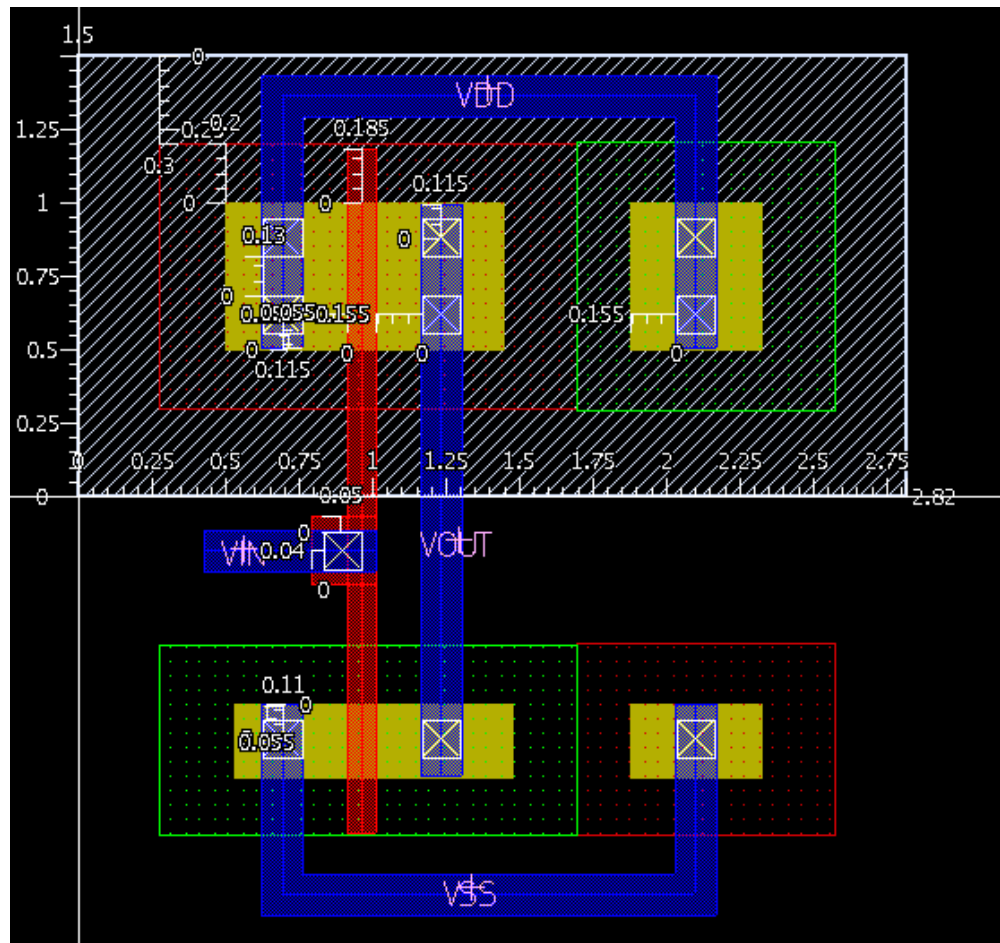


Figure 1: Inverter Layout

1.2 DRC Result (CLEAN)

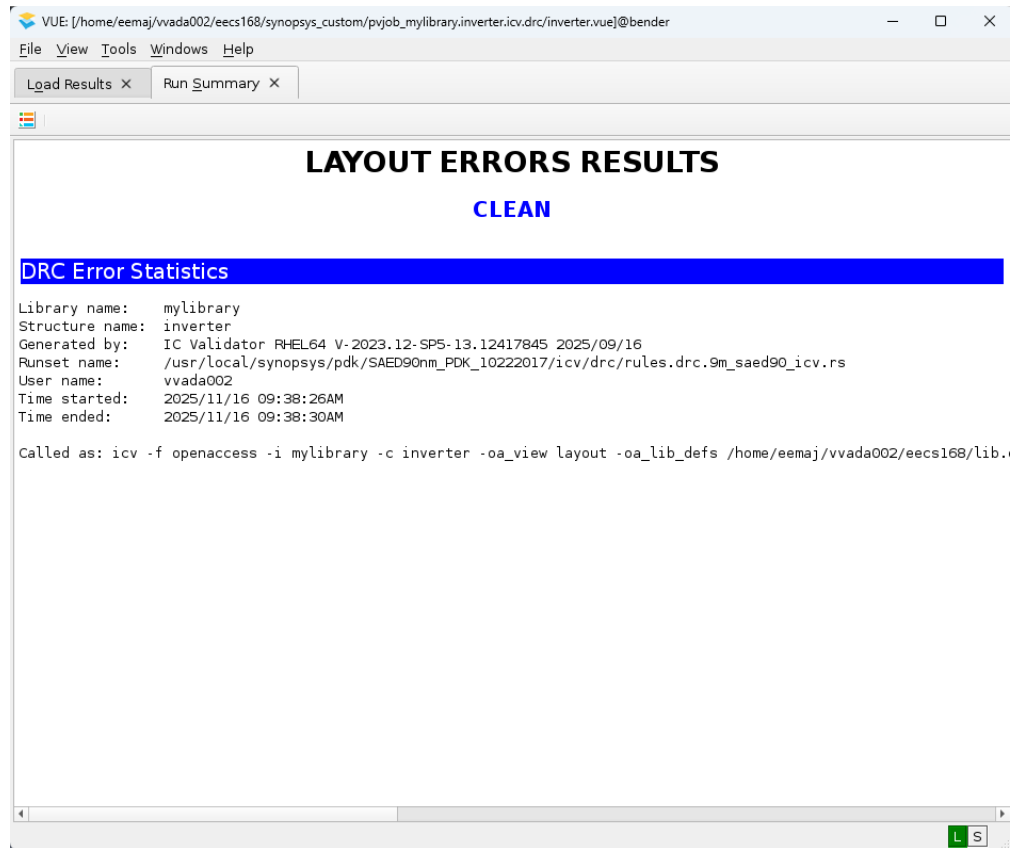


Figure 2: DRC: CLEAN

2 NAND Gate: Schematic, Simulation, Layout

2.1 NAND Schematic

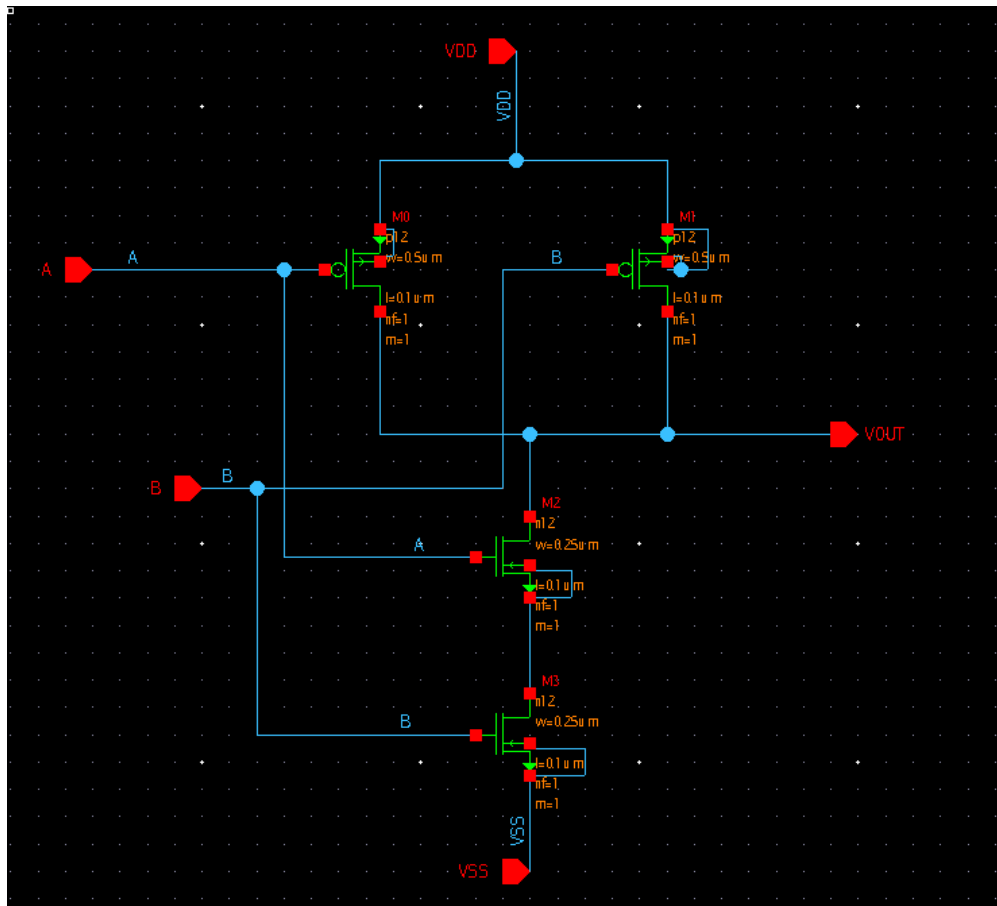


Figure 3: NAND Gate Schematic

2.2 NAND Testbench

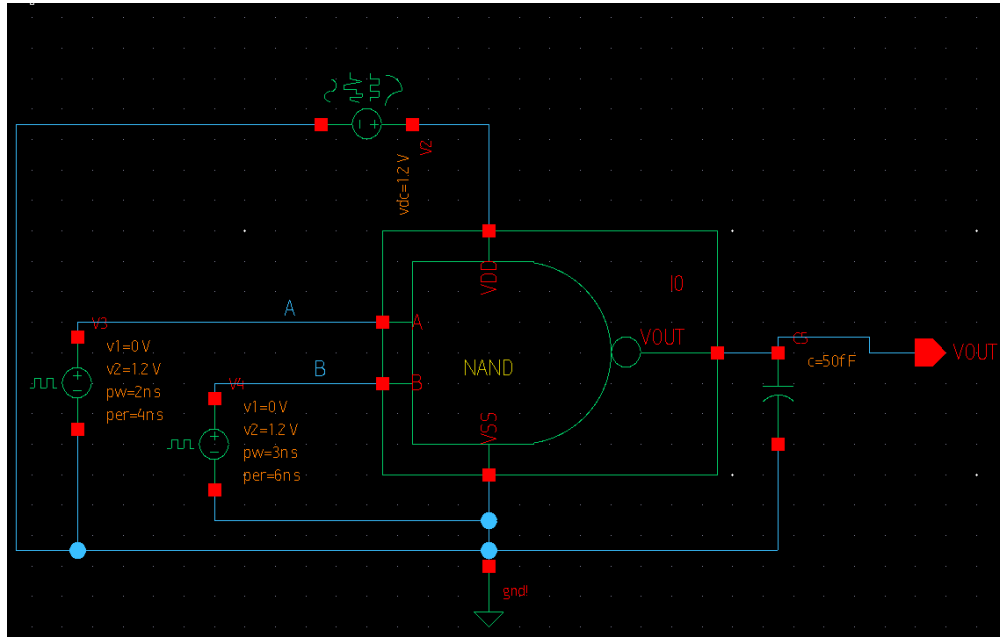


Figure 4: NAND Testbench Schematic

2.3 NAND DC Simulation Waveform

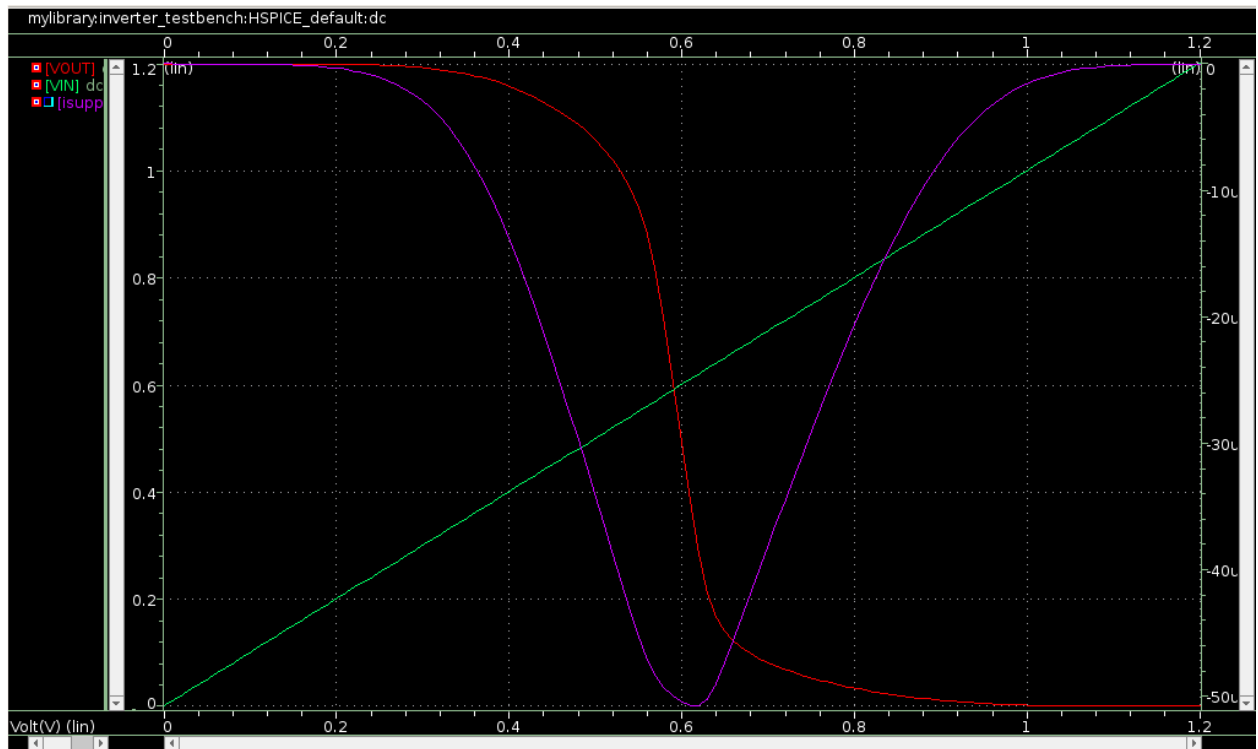


Figure 5: DC Transient Simulation Waveform for NAND Gate

2.4 NAND Trans Simulation Waveform

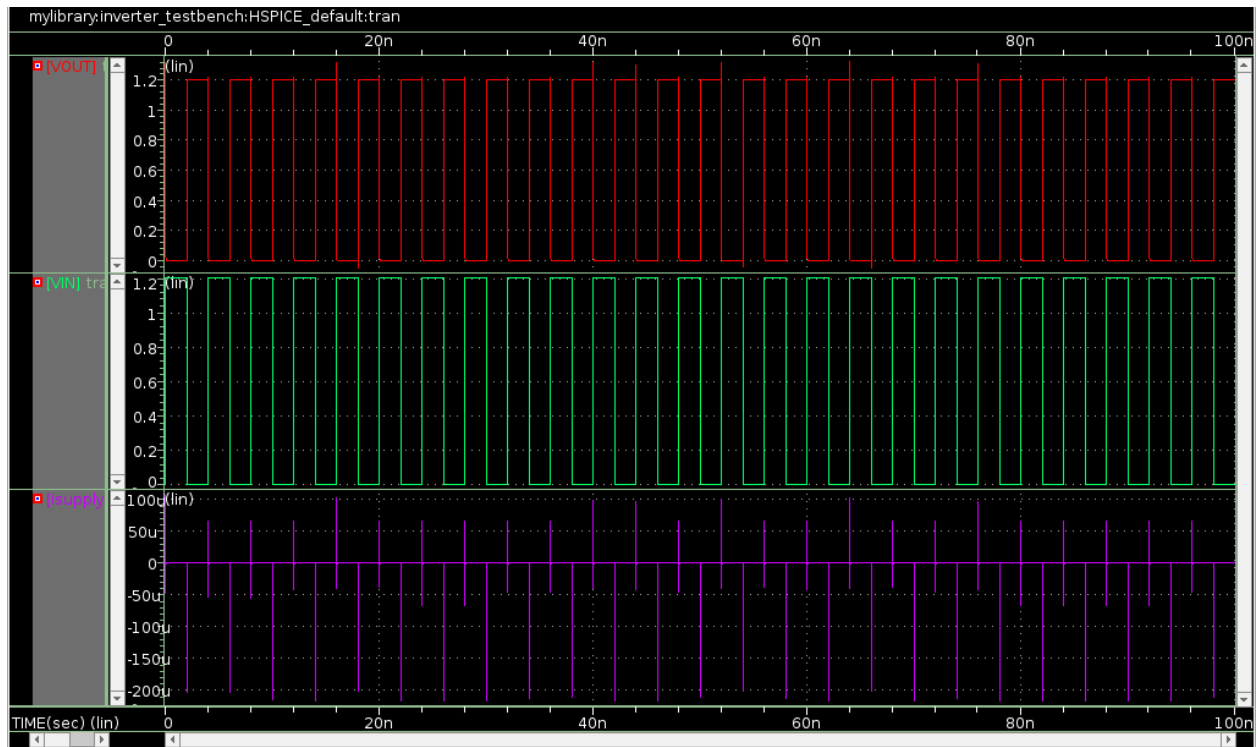


Figure 6: TRANS Transient Simulation Waveform for NAND Gate

2.5 NAND Layout

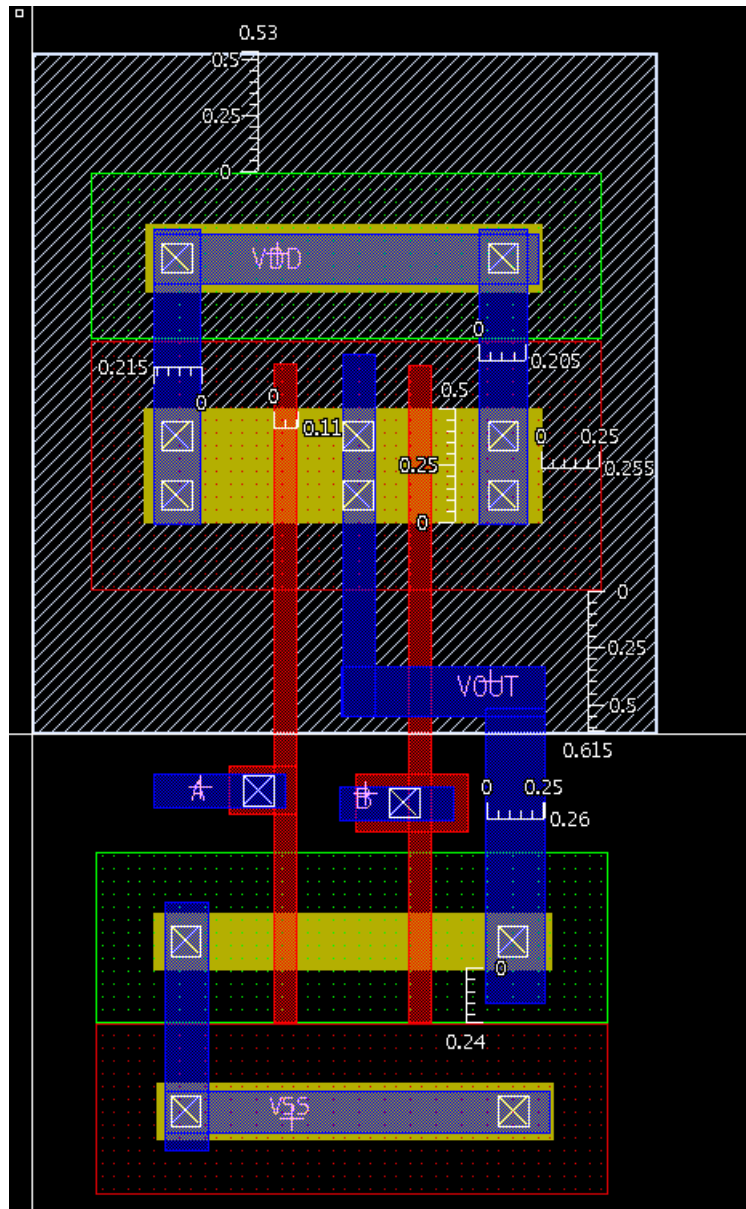


Figure 7: NAND Gate Layout

2.6 DRC Result (CLEAN)

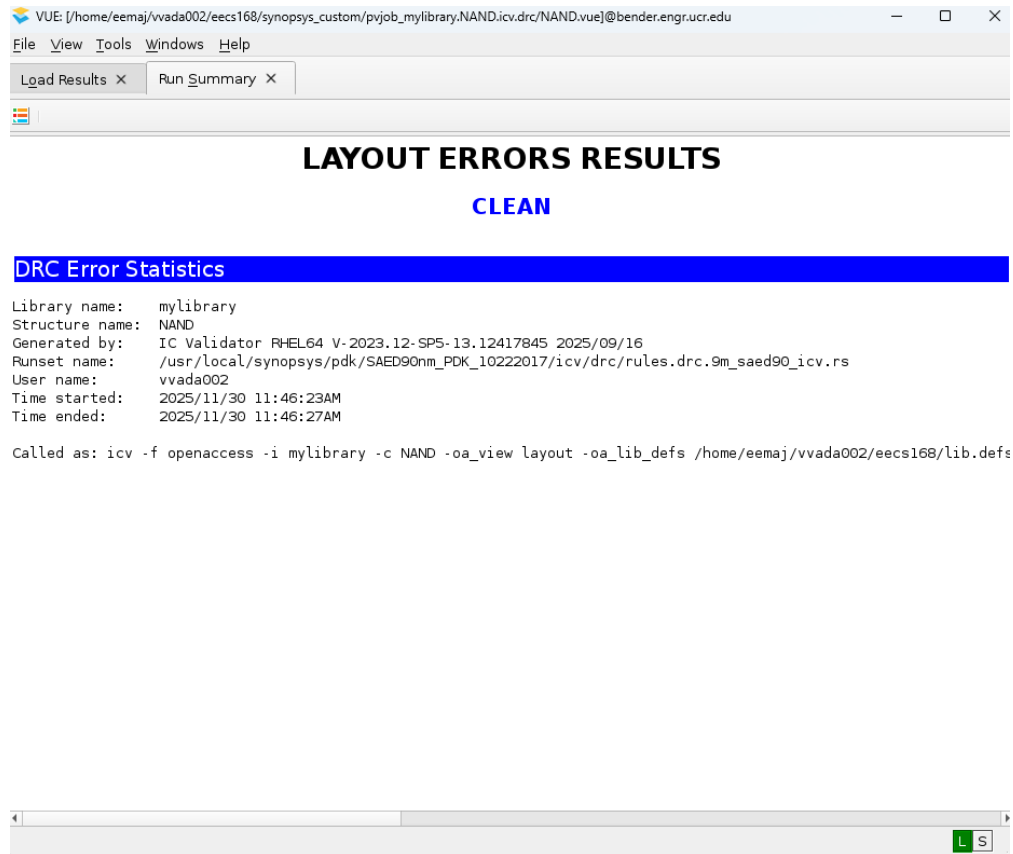


Figure 8: DRC: CLEAN (NAND)

3 Ring Oscillator (Hierarchical Design)

3.1 Ring Oscillator Schematic

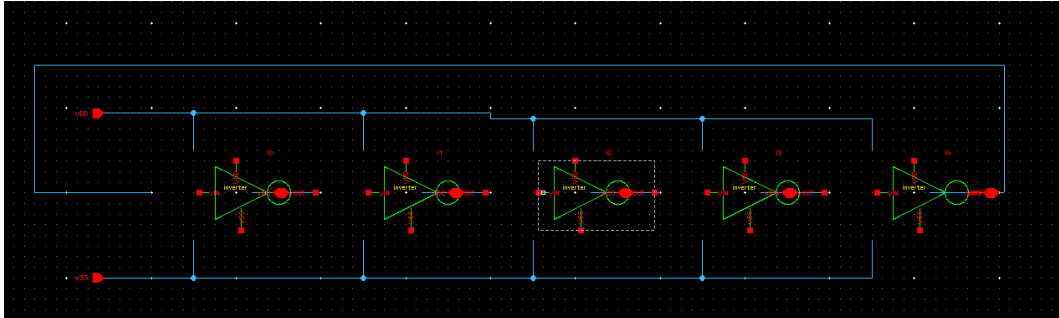


Figure 9: Ring Oscillator Schematic

3.2 Ring Oscillator Layout

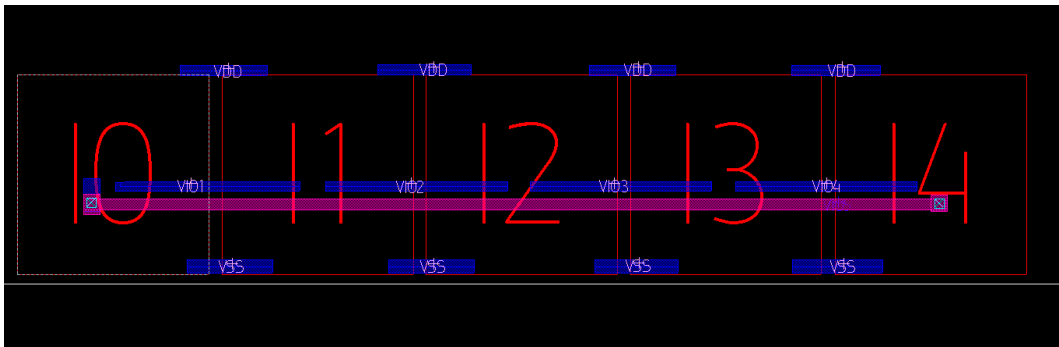


Figure 10: Ring Oscillator Layout

3.3 Ring Oscillator Testbench

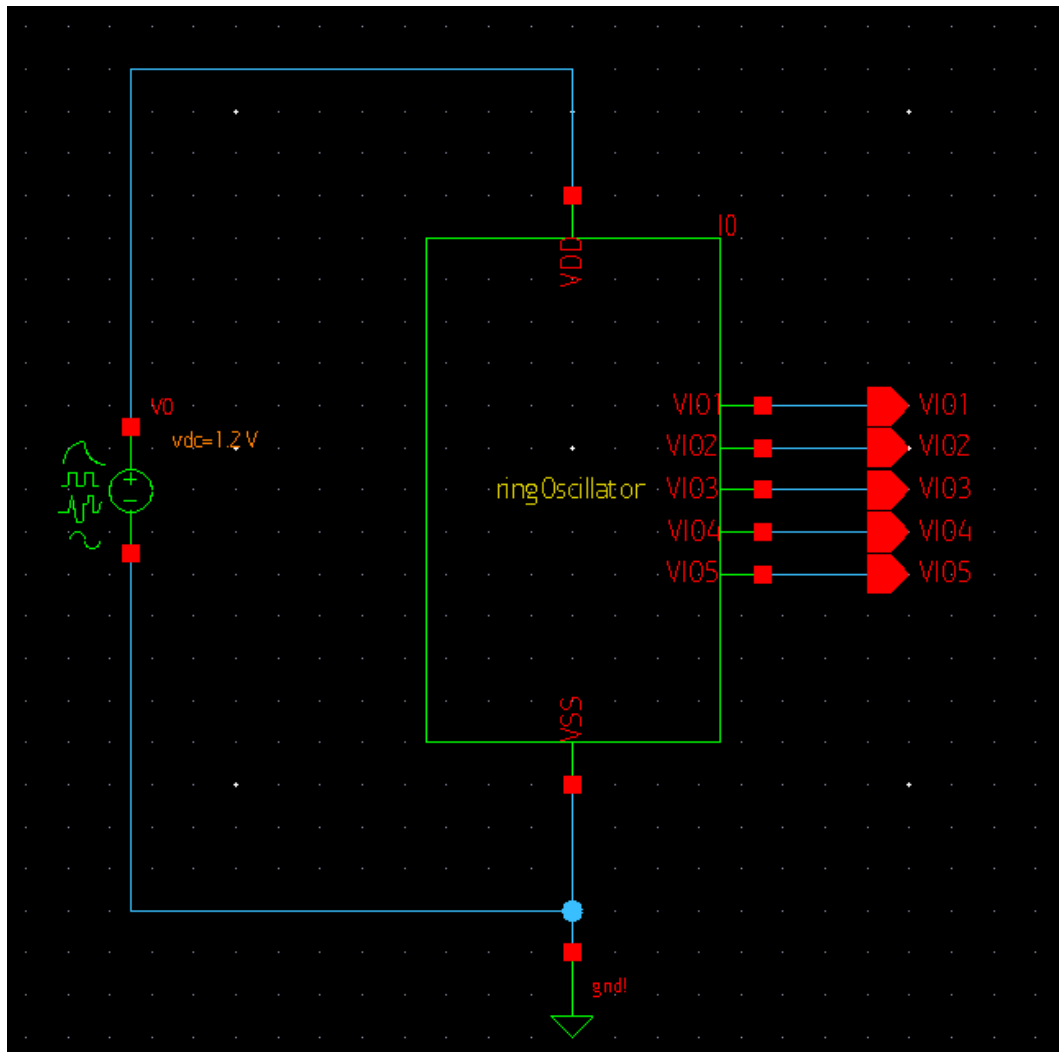


Figure 11: Ring Oscillator Testbench

3.4 DRC Result (CLEAN)

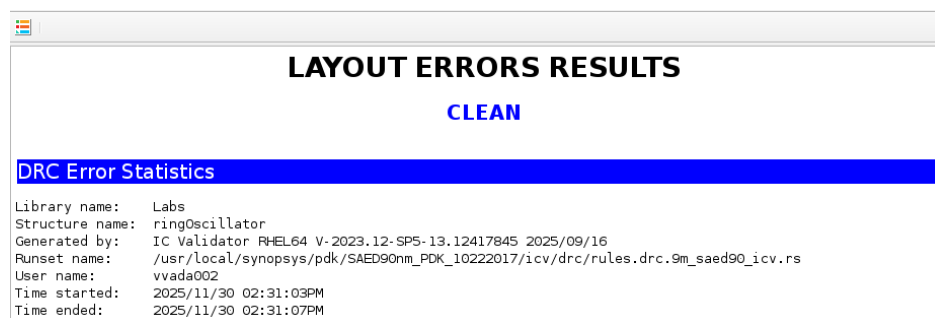


Figure 12: DRC: CLEAN (Ring Oscillator)

4 Issues Encountered

Some of the main difficulties I ran into during this lab came from the layout tools themselves rather than from the CMOS concepts. When I started drawing the inverter and later the NAND gate, certain layers in the PDK wouldn't display their usual colors, fill patterns, or material properties. This made it unexpectedly hard to distinguish between diffusion, poly, metal, contacts, and wells while building the layout. The visual cues normally help you check alignment, enclosure, overlaps, and spacing at a glance, so losing that color information slowed down my workflow and forced me to keep re-checking layer selections manually. It also caused a few mistakes, like placing the wrong type of contact or accidentally selecting the wrong layer for routing, because everything looked too similar.

On top of the display problems, I also ran into verification issues that pushed the lab further out. Even after the shapes looked correct, the tool flagged DRC violations that didn't seem to match what I had drawn. Some of these came from tiny spacing constraints or enclosure rules that I didn't realize I had violated — things like diffusion not extending far enough under a contact, or metal spacing being off by only a fraction of a lambda. These small errors can be hard to spot until DRC catches them.

Although we didn't do LVS in this lab, the process of repeatedly debugging DRC and fixing layout mistakes still introduced delays and made me think more carefully about how each device is actually constructed in the layout. Every time DRC flagged something, it forced me to trace back what rule I broke and why, which gave me a better understanding of the physical requirements behind the inverter and NAND structures.

Overall, the technical issues with layer visibility and verification slowed my progress, but they also made the layout process feel more realistic — since in real chip design, tools don't always behave perfectly, and tiny geometric details matter a lot more than they do in schematics.